

Arturia Modes

This guide describes the Arturia KeyLab Essential 61 mk3 modes used with the Pedro performance rig. The complete Performance Rig reference is at [src/performance-rigs/pedro-performance-rig/README.md](#).

The Arturia pad selector is a Linux user-session service. It does not require the Arturia Windows editor.

Start And Stop

The selector service is installed on `airstar`, but it is not enabled at login.

```
systemctl --user start music-rig-arturia-profile-session.service
systemctl --user stop music-rig-arturia-profile-session.service
```

Stopping the service restores `full-live-rack` and removes temporary candidate connections.

Pad Modes

Pad management matches MIDI channel 10 only. Normal keyboard notes use channel 1 and are not interpreted as mode changes.

Pad	Mode
Bank A Pad 1	Tonewheel Organ
Bank A Pad 2	Synth Programmer
Bank A Pad 3	Full Live Rack
Bank A Pad 4	Worship Piano
Bank A Pad 5	Gospel Keys
Bank A Pad 6	Ambient Worship
Bank A Pad 7	Jazz Keys
Bank A Pad 8	Soul and R&B
Bank B Pad 1	Cinematic Strings
Bank B Pad 2	Orchestral
Bank B Pad 3	Brass and Winds

Pad	Mode
Bank B Pad 4	Synthwave
Bank B Pad 5	Retro Keys
Bank B Pad 6	Intimate Pads
Bank B Pad 7	Praise Leads
Bank B Pad 8	Acoustic Worship

Full Live Rack

This is the protected default and recovery mode.

Fader	Instrument volume	Knob	Instrument reverb
1	Basic Piano	1	Basic Piano
2	Nord White Grand	2	Nord White Grand
3	Alt Strings	3	Alt Strings
4	Good Flute	4	Good Flute
5	SAX Lirakeys	5	SAX Lirakeys
6	Hammond Organ Fast	6	Hammond Organ Fast
7	Optik Synth	7	Optik Synth
8	PAD EFEITOS	8	PAD EFEITOS
9	AtmosferaPAD	9	AtmosferaPAD

Synth Programmer

Faders use pickup behavior: the value changes after the physical control crosses the stored value.

Control	Parameter
Fader 1	Oscillator 1 mix, <code>DC01_BALANCE</code>
Fader 2	Oscillator 2 level/balance, <code>DC02_BALANCE</code>
Fader 3	Filter cutoff, <code>DCF1_CUTOFF</code>
Fader 4	Filter resonance, <code>DCF1_RES0</code>

Control	Parameter
Fader 5	Amp attack, <code>DCA1_ATTACK</code>
Fader 6	Amp release, <code>DCA1_RELEASE</code>
Fader 7	Modulation depth, <code>LF01_BALANCE</code>
Fader 8	Modulation rate, <code>LF01_RATE</code>
Fader 9	Unison/detune
Knob 1	Drive/compression
Knob 2	Effects send/mix
Knob 3	Delay time
Knob 4	Delay feedback
Knob 5	Reverb size
Knob 6	Reverb mix
Knob 7	Pitch modulation
Knob 8	Synth volume, <code>OUT1_VOLUME</code>
Knob 9	Synth reverb width

Tonewheel Organ

The organ uses setBfree. Drawbars are inverted like a physical organ: down is loudest and up is off.

Fader	Drawbar	Knob	Function
1	16'	1	Percussion enable/mode
2	5 1/3'	2	Percussion decay
3	8'	3	Overdrive input
4	4'	4	Vibrato/chorus selection
5	2 2/3'	5	Rotary stop/slow/fast
6	2'	6	Overdrive output
7	1 3/5'	7	Vibrato routing
8	1 1/3'	8	Swell volume
9	1'	9	Reverb mix

Materialized Genre Modes

Pads 4 through 16 load separate headless Carla projects. Each project has nine real SoundFont or DecentSampler patches from the larger library. These are not the nine live-rack instruments. The source paths are recorded in [benchmarks/arturia-genre-patches.json](#).

In every genre mode, Faders 1-9 control the volume of the nine patches below and Knobs 1-9 control their corresponding reverb. The central encoder remains Master volume and its click remains Master mute.

Mode	Faders 1 through 9, in order
Worship Piano	Wurlitzer, CFX Grand, Slinky Violin Duet, JF Gospel Organ, Cinematic Strings, JF Worship Piano, Sky Pad Evolving, Endless Church Chime, Worship Guitar
Gospel Keys	Wurlitzer, Motif ES6 Piano, Slinky Violin Duet, JF Gospel Organ, Alto Sax, Dark Violins, Choir Korg Aahhs, Laboriel Slap Bass, Shiny Effects Pad
Ambient Worship	Sky Pad Evolving, CFX Grand, Subfrost Harmonic Bow, Cinematic Strings, Shiny Effects Pad, CS-20M Big Waves, JF Worship Piano, Endless Church Chime, Choir Korg Aahhs
Jazz Keys	Wurlitzer, Rhodes VS Extreme, Slinky Violin Duet, Motif ES6 Piano, Alto Sax, Dark Violins, Worship Guitar, Short Scale Bass, Endless Church Chime
Soul and R&B	Wurlitzer, Rhodes VS Extreme, Slinky Violin Duet, JF Worship Piano, Alto Sax, Laboriel Slap Bass, Worship Guitar, Dark Violins, Shiny Effects Pad
Cinematic Strings	Sky Pad Evolving, Dark Violins, Subfrost Harmonic Bow, CFX Grand, Cinematic Strings, Choir Korg Aahhs, Endless Church Chime, D-50 Stack, Motif ES6 Piano
Orchestral	Box Harp Picked, Cinematic Strings, Slinky Violin Duet, Alto Sax, Mariachi Trumpet, Choir Korg Aahhs, Endless Church Chime, CFX Grand, Subfrost Harmonic Bow
Brass and Winds	Wurlitzer, Alto Sax, CS-20M Big Waves, Motif ES6 Piano, Worship Guitar, Laboriel Slap Bass, Dark Violins, Endless Church Chime, Mariachi Trumpet
Synthwave	CS-20M Big Waves, D-50 Stack, Sky Pad Evolving, Rhodes VS Extreme, Wurlitzer, Shiny Effects Pad, Motif ES6 Piano, Laboriel Slap Bass, Endless Church Chime
Retro Keys	Wurlitzer, Rhodes VS Extreme, CS-20M Big Waves, Motif ES6 Piano, Hammond B3 Slow, JF Super Saw, Laboriel Slap Bass, Shiny Effects Pad, Endless Church Chime
Intimate Pads	Wurlitzer, CFX Grand, Slinky Violin Duet, Dark Violins, Sky Pad Evolving, Subfrost Harmonic Bow, Box Harp Picked, Endless Church Chime, Choir Korg Aahhs
Praise Leads	CS-20M Big Waves, JF Gospel Organ, Sky Pad Evolving, D-50 Stack, JF Worship Piano, Cinematic Strings, Shiny Effects Pad, Choir Korg Aahhs, Endless Church Chime
Acoustic Worship	Wurlitzer, CFX Grand, Slinky Violin Duet, Dark Violins, Worship Guitar, JF Gospel Organ, Alto Sax, Box Harp Picked, Sky Pad Evolving

SMK-25, SMC-Mixer, SMC-PAD, and SMC-PAD Pocket are not part of these Arturia mode changes. Their existing MIDI, audio, pad, transport, and mixer behavior remains active.

Headless Carla

The protected full rack can also run as an opt-in headless Carla user service. Close normal GUI Carla first.

```
docs/tools/music-rig/packaging/linux/install-pedro-carla-headless
systemctl --user start music-rig-pedro-carla-headless.service
~/.local/bin/carla-pedro-osc-gui
```

The OSC GUI attaches to the running backend; it does not launch a second Carla engine. Stop the backend before reopening normal GUI Carla:

```
systemctl --user stop music-rig-pedro-carla-headless.service
```

Safety

- `full-live-rack` remains the default and rollback mode.
- Candidate engines are user-owned and are not installed system-wide by the

session installer.

- The service is manually started and can be stopped immediately.
- The protected Carla project and current live services are not replaced by a

pad mode.

PEDRO PERFORMANCE RIG

Pedro Performance Rig

This directory contains the portable authored definition of the first Performance Rig. Schema version `music-studies/performance-rig/v1` uses JSON Schema Draft 2020-12.

Current status: schema, stable-slot, Hardware Preset, Device Profile, Rig Profile, Linux Platform Binding extraction, deterministic compilation envelope, pad triggers, mapped synthv1/setBfree candidates, and materialized genre projects. The authored files remain the source of truth for candidate selection; the protected `setup.json`, Carla project, services, and graph remain the production authority and default recovery state.

`rig.json` records the five stable controller slots from the protected rack. Each slot orders selectors from model, semantic alias, and endpoint purpose to an optional local discriminator and optional USB ID. The four M-VAVE devices share USB ID `4353:4b4d`, so that ID is supporting evidence and cannot distinguish them by itself. Platform bindings resolve these portable selectors to operating-system device identities.

All six Hardware Preset IDs resolve to verified files in `hardware-presets`. The SMC-PAD and SMC-PAD Pocket assignments are checked against the 2026-08-11 live capture, including exact per-pad channel/note assignments and the Pocket's eight hardware-internal control pads.

The current Device Profiles resolve under `device-profiles/<slot>/<id>`:

- Arturia `multi-instrument-rack`;
- SMK-25 `ambient-pad-layers`;
- SMC-Mixer `eight-band-eq`;
- SMC-PAD `drum-set`; and
- SMC-PAD Pocket `drum-set`.

S2 adds Arturia candidate profiles:

- `tonewheel-organ`, with nine drawbar mappings and organ controls;
- `tonewheel-organ-setbfree`, with the same Arturia layout and a setBfree MIDI backend;
- `synth-programmer`, with oscillator, filter, envelope, modulation, and effects mappings;
- `synth-programmer-synthv1`, with a native synthv1 controller map; and
- thirteen genre profiles using materialized SoundFont/DecentSampler patch sets.

The setBfree and synthv1 candidates use their own headless engines and the verified Arturia hardware control surface. Genre profiles use separate candidate Carla projects generated from `docs/tools/music-rig/benchmarks/arturia-genre-patches.json`. All candidates remain explicit, reversible sessions; the protected live Carla project is not replaced.

They link semantic mappings to controls from the selected Hardware Preset and declare capabilities, ownership, dependencies, readiness, state, takeover, and switch safety without platform paths or backend identifiers. The two pad profiles intentionally share `drum-set.notes`; other current ownership remains exclusive or read-only.

`rig-profiles/full-live-rack.json` composes all five roles as the default and fallback Rig Profile. It declares the aggregate endpoint and musical capabilities, pins the eighteen current sound engines, identifies the shared drum engine and effects, and retains the safe takeover and rollback policies. It is authored data only; the protected setup remains the active default. The resolved `switch-triggers.json` management catalogue maps the 16 Arturia pads to the organ, synth, live, and genre profiles.

`rig-profiles/tonewheel-organ.json`, `rig-profiles/tonewheel-organ-setbfree.json`, and the synth/genre profiles compose Arturia alternatives with the unchanged SMK-25, SMC-Mixer, SMC-PAD, and Pocket roles. `full-live-rack` remains the protected default and fallback.

The seven schemas are:

- `common.schema.json`: shared identifiers, selectors, capabilities, MIDI,

ownership, readiness, takeover, state, and switch-safety types;

- `rig.schema.json`: the complete Rig catalogue and stable device slots;
- `rig-profile.schema.json`: one global composition;
- `device-profile.schema.json`: one role for one logical device slot;
- `hardware-preset.schema.json`: controller-local raw message assignments;
- `platform-binding.schema.json`: backend device identities, semantic target

locators, machine paths, lifecycle resources, and evidence status; and

- `switch-triggers.schema.json`: persistent management events translated to

the same future switch operations as the CLI.

Top-level authored definitions contain semantic capabilities. Operating-system paths, PipeWire/JACK port names, Carla parameter indices, Windows device IDs, and service identifiers belong only in Platform Bindings. The authored `airstar-current` Linux binding resolves the complete `full-live-rack` contract against the protected setup. It is authoring-only and cannot mutate or activate the runtime. The Windows document under the validator fixtures proves the portable contract shape; it is explicitly not physical Windows support or certification.

Root validation checks the Rig, all Hardware Presets, Device Profiles, Rig Profiles, Platform Bindings, and Switch Triggers. It resolves slot, model, endpoint, preset, control, profile, capability, resource, binding, trigger source, and trigger-operation references; checks required-slot coverage, aggregate capabilities and readiness, pinned and shared resources, initial-state ownership, explicit composition ownership conflicts, management MIDI assignment, readiness ceilings, and consumed-event mapping conflicts; and locks the current Linux binding to protected Airstar aliases, paths, checksums, and services. It reads authored files and protected evidence only; it does not connect to the live rig.

Run the offline schema suite from the repository root after installing the authoring-only dependency:

```
python3 -m pip install -r docs/tools/music-rig/requirements-schema.txt
python3 docs/tools/music-rig/validate-performance-rig.py --self-test
python3 docs/tools/music-rig/validate-performance-rig.py \
  --validate-root src/performance-rigs/pedro-performance-rig \
  --authority-setup docs/tools/airstar-live-setup/setup.json \
  --authority-pad-capture \
    docs/tools/music-rig/benchmarks/hardware-preset-airstar-2026-08-11.json

python3 docs/tools/music-rig/compile-performance-rig.py \
  --rig-root src/performance-rigs/pedro-performance-rig \
  --platform-binding airstar-current \
  --check-only
```

The compiler emits only to an explicit output path and refuses to overwrite authored source. Its Milestone 2 lookup, ownership, graph-delta, and fingerprint contracts are documented in [COMPILER.md](#) . It does not materialize or activate Carla, Patchbay, MIDI, audio, service, or runtime state.

Musician Guide

The Arturia KeyLab Essential 61 mk3 can select the following modes from its 16 pads. Bank A and Bank B each contain eight pads. Pad management uses MIDI channel 10; normal keyboard notes use channel 1 and are not mode triggers.

Pad	Mode
Bank A 1	Tonewheel Organ (tonewheel-organ-setbfree)
Bank A 2	Synth Programmer (synth-programmer-synthv1)
Bank A 3	Full Live Rack (full-live-rack)
Bank A 4	Worship Piano (worship-piano)
Bank A 5	Gospel Keys (gospel-keys)
Bank A 6	Ambient Worship (ambient-worship)
Bank A 7	Jazz Keys (jazz-keys)
Bank A 8	Soul and R&B (soul-rnb)
Bank B 1	Cinematic Strings (cinematic-strings)
Bank B 2	Orchestral (orchestral)
Bank B 3	Brass and Winds (brass-winds)
Bank B 4	Synthwave (synthwave)
Bank B 5	Retro Keys (retro-keys)
Bank B 6	Intimate Pads (intimate-pads)

Pad	Mode
Bank B 7	Praise Leads (praise-leads)
Bank B 8	Acoustic Worship (acoustic-worship)

Start the explicit Linux user session on [airstar](#) before playing:

```
systemctl --user start music-rig-arturia-profile-session.service
```

Stop it after playing. This restores [full-live-rack](#) and removes candidate processes and links:

```
systemctl --user stop music-rig-arturia-profile-session.service
```

The central Arturia encoder is Master volume in every mode. Pressing it toggles Master mute. The faders and knobs below use the same physical positions in every genre project: each fader controls the volume of its listed patch, and the knob below it controls that patch's reverb.

Full Live Rack

This is the protected current default and recovery mode.

Fader	Instrument	Knob	Function
1	Basic Piano	1	Basic Piano reverb
2	Nord White Grand	2	Nord White Grand reverb
3	Alt Strings	3	Alt Strings reverb
4	Good Flute	4	Good Flute reverb
5	SAX Lirakeys	5	SAX Lirakeys reverb
6	Hammond Organ Fast	6	Hammond Organ Fast reverb
7	Optik Synth	7	Optik Synth reverb
8	PAD EFEITOS	8	PAD EFEITOS reverb
9	AtmosferaPAD	9	AtmosferaPAD reverb

Synth Programmer

Control	Function
Fader 1	Oscillator 1 mix (DC01_BALANCE)

Control	Function
Fader 2	Oscillator 2 level/balance (DC02_BALANCE)
Fader 3	Filter cutoff (DCF1_CUTOFF)
Fader 4	Filter resonance (DCF1_RES0)
Fader 5	Amp attack (DCA1_ATTACK)
Fader 6	Amp release (DCA1_RELEASE)
Fader 7	Modulation depth (LF01_BALANCE)
Fader 8	Modulation rate (LF01_RATE)
Fader 9	Unison/detune
Knob 1	Drive/compression
Knob 2	Effects send/mix
Knob 3	Delay time
Knob 4	Delay feedback
Knob 5	Reverb size
Knob 6	Reverb mix
Knob 7	Pitch modulation
Knob 8	Synth volume (OUT1_VOLUME)
Knob 9	Synth reverb width

Tonewheel Organ

Fader	Drawbar	Knob	Function
1	16'	1	Percussion enable/mode
2	5 1/3'	2	Percussion decay
3	8'	3	Overdrive input
4	4'	4	Vibrato/chorus selection
5	2 2/3'	5	Rotary stop/slow/fast
6	2'	6	Overdrive output
7	1 3/5'	7	Vibrato routing

Fader	Drawbar	Knob	Function
8	1 1/3'	8	Swell volume
9	1'	9	Reverb mix

Drawbars are inverted like a physical organ: down is loudest and up is off.

Genre Patch Sets

Pads 4 through 16 use materialized Carla projects built from the installed SoundFont and DecentSampler libraries. The exact patch source paths are recorded in `arturia-genre-patches.json`.

Mode	Faders 1 through 9, in order
Worship Piano	Wurlitzer, CFX Grand, Slinky Violin Duet, JF Gospel Organ, Cinematic Strings, JF Worship Piano, Sky Pad Evolving, Endless Church Chime, Worship Guitar
Gospel Keys	Wurlitzer, Motif ES6 Piano, Slinky Violin Duet, JF Gospel Organ, Alto Sax, Dark Violins, Choir Korg Aahhs, Laboriel Slap Bass, Shiny Effects Pad
Ambient Worship	Sky Pad Evolving, CFX Grand, Subfrost Harmonic Bow, Cinematic Strings, Shiny Effects Pad, CS-20M Big Waves, JF Worship Piano, Endless Church Chime, Choir Korg Aahhs
Jazz Keys	Wurlitzer, Rhodes VS Extreme, Slinky Violin Duet, Motif ES6 Piano, Alto Sax, Dark Violins, Worship Guitar, Short Scale Bass, Endless Church Chime
Soul and R&B	Wurlitzer, Rhodes VS Extreme, Slinky Violin Duet, JF Worship Piano, Alto Sax, Laboriel Slap Bass, Worship Guitar, Dark Violins, Shiny Effects Pad
Cinematic Strings	Sky Pad Evolving, Dark Violins, Subfrost Harmonic Bow, CFX Grand, Cinematic Strings, Choir Korg Aahhs, Endless Church Chime, D-50 Stack, Motif ES6 Piano
Orchestral	Box Harp Picked, Cinematic Strings, Slinky Violin Duet, Alto Sax, Mariachi Trumpet, Choir Korg Aahhs, Endless Church Chime, CFX Grand, Subfrost Harmonic Bow
Brass and Winds	Wurlitzer, Alto Sax, CS-20M Big Waves, Motif ES6 Piano, Worship Guitar, Laboriel Slap Bass, Dark Violins, Endless Church Chime, Mariachi Trumpet
Synthwave	CS-20M Big Waves, D-50 Stack, Sky Pad Evolving, Rhodes VS Extreme, Wurlitzer, Shiny Effects Pad, Motif ES6 Piano, Laboriel Slap Bass, Endless Church Chime
Retro Keys	Wurlitzer, Rhodes VS Extreme, CS-20M Big Waves, Motif ES6 Piano, Hammond B3 Slow, JF Super Saw, Laboriel Slap Bass, Shiny Effects Pad, Endless Church Chime
Intimate Pads	Wurlitzer, CFX Grand, Slinky Violin Duet, Dark Violins, Sky Pad Evolving, Subfrost Harmonic Bow, Box Harp Picked, Endless Church Chime, Choir Korg Aahhs
Praise Leads	CS-20M Big Waves, JF Gospel Organ, Sky Pad Evolving, D-50 Stack, JF Worship Piano, Cinematic Strings, Shiny Effects Pad, Choir Korg Aahhs, Endless Church Chime
Acoustic Worship	Wurlitzer, CFX Grand, Slinky Violin Duet, Dark Violins, Worship Guitar, JF Gospel Organ, Alto Sax, Box Harp Picked, Sky Pad Evolving

The SMK-25, SMC-Mixer, SMC-PAD, and SMC-PAD Pocket are not part of these Arturia mode changes. Their existing MIDI, audio, pad, transport, and mixer behavior remains active.

Headless Carla

The protected live rack and genre projects can run headlessly. To install the optional headless full-rack service:

```
docs/tools/music-rig/packaging/linux/install-pedro-carla-headless
```

Close normal GUI Carla first, then start the headless backend:

```
systemctl --user start music-rig-pedro-carla-headless.service
```

To attach the optional Carla control GUI through OSC:

```
~/local/bin/carla-pedro-osc-gui
```

Do not open the same project in a second normal Carla instance. Stop the headless service before returning to GUI Carla:

```
systemctl --user stop music-rig-pedro-carla-headless.service
```